Abstract linear algebra homework10

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Problem 1

Let V be a finite dimensional real inner product space.

(a)

Let $W \subseteq \mathcal{L}(V)$ denote the set of self-adjoint operators on V, i.e.

$$W = \{\, T \in \mathcal{L}(V) : T = T^* \,\}.$$

We show W is a subspace of $\mathcal{L}(V)$.

- **Zero vector.** The zero operator $0 \in \mathcal{L}(V)$ satisfies $\langle 0v, w\rangle = 0 = \langle v, 0w\rangle$ for all $v, w \in V$, so $0^* = 0$. Hence $0 \in W$.
- **Closed under addition.** Suppose $S, T \in W$, so $S^* = S$ and $T^* = T$. Recall the adjoint satisfies $(S+T)^* = S^* + T^*$ (this follows since for all v, w,

  $$\langle (S+T)v, w\rangle = \langle Sv, w\rangle + \langle Tv, w\rangle = \langle v, S^*w\rangle + \langle v, T^*w\rangle = \langle v, (S^*+T^*)w\rangle,$$

  which shows $(S+T)^* = S^* + T^*$ by uniqueness of the adjoint). Hence

  $$(S+T)^* = S^* + T^* = S + T,$$

  so $S + T \in W$.
- **Closed under scalar multiplication.** Suppose $T \in W$ and $c \in \mathbb{R}$. For all $v, w \in V$,

  $$\langle (cT)v, w\rangle = c\langle Tv, w\rangle = c\langle v, T^*w\rangle = \langle v, cT^*w\rangle,$$

  (using that c is real, so it can be pulled through either slot of the inner product), which shows $(cT)^* = cT^*$. Hence

  $$(cT)^* = cT^* = cT,$$

  so $cT \in W$.

Since W contains 0 and is closed under addition and scalar multiplication, W is a subspace of $\mathcal{L}(V)$.

(b)

Let $n = \dim(V)$. Fix an orthonormal basis of V; with respect to this basis, an operator $T \in \mathcal{L}(V)$ is self-adjoint if and only if its matrix A is symmetric, i.e. $A = A^T$ (this is the standard correspondence between the adjoint of an operator and the conjugate transpose of its matrix in an orthonormal basis; over $\mathbb{R}$ this is just the transpose). So W is isomorphic, as a vector space, to the space of $n \times n$ real symmetric matrices.

A symmetric $n \times n$ matrix is determined freely by its entries on and above the diagonal: the diagonal has n free entries, and the strictly upper-triangular part has $\binom{n}{2} = \frac{n(n-1)}{2}$ free entries (each determining the corresponding lower entry by symmetry). Hence

$$\dim(W) = n + \binom{n}{2} = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

So the dimension of the subspace of self-adjoint operators is $\dfrac{\dim(V)(\dim(V)+1)}{2}$

Problem 2

Let V be a finite dimensional real inner product space and $T : V \to V$ self-adjoint. By the Real Spectral Theorem, T has an orthonormal eigenbasis $e_1, \ldots, e_n$ of V, with corresponding (real) eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, i.e. $Te_i = \lambda_i e_i$.

Let $c \in \mathbb{R}$, $\varepsilon > 0$, and suppose $v \in V$ has $\|v\| = 1$ and $\|Tv - cv\| < \varepsilon$. Write

$$v = \sum_{i=1}^{n} a_i e_i, \qquad a_i \in \mathbb{R}.$$

Since $\{e_i\}$ is orthonormal, $\|v\|^2 = \sum_i a_i^2 = 1$.

Now,

$$Tv - cv = \sum_{i=1}^{n} a_i \lambda_i e_i - \sum_{i=1}^{n} a_i c\, e_i = \sum_{i=1}^{n} a_i (\lambda_i - c)\, e_i.$$

Since $\{e_i\}$ is an orthonormal basis, the (generalized) Pythagorean theorem gives

$$\|Tv - cv\|^2 = \sum_{i=1}^{n} a_i^2 (\lambda_i - c)^2.$$

Suppose, for contradiction, that $|\lambda_i - c| \geq \varepsilon$ for every $i = 1, \ldots, n$. Then $(\lambda_i - c)^2 \geq \varepsilon^2$ for all i, so

$$\|Tv - cv\|^2 = \sum_{i=1}^{n} a_i^2 (\lambda_i - c)^2 \geq \varepsilon^2 \sum_{i=1}^{n} a_i^2 = \varepsilon^2 \|v\|^2 = \varepsilon^2.$$

But this contradicts $\|Tv - cv\| < \varepsilon$, i.e. $\|Tv - cv\|^2 < \varepsilon^2$.

Hence there must exist some index i with $|\lambda_i - c| < \varepsilon$. Taking $\lambda = \lambda_i$, we have found an eigenvalue λ of T with $|c - \lambda| < \varepsilon$, as desired. ■

Problem 3

Let V be a finite dimensional real inner product space, $T : V \to V$ self-adjoint, and let $S : V \to V$ be a linear map such that every eigenvector of T is also an eigenvector of S. We show $ST = TS$.

By the Real Spectral Theorem, since T is self-adjoint, V has a basis $e_1, \dots, e_n$ consisting of eigenvectors of T, say $Te_i = \lambda_i e_i$ for real numbers λ_i.

By hypothesis, each e_i (being an eigenvector of T) is also an eigenvector of S: there exist scalars μ_i such that $Se_i = \mu_i e_i$.

Now compute, for each i:

$$(ST)(e_i) = S(Te_i) = S(\lambda_i e_i) = \lambda_i S(e_i) = \lambda_i \mu_i e_i,$$

$$(TS)(e_i) = T(Se_i) = T(\mu_i e_i) = \mu_i T(e_i) = \mu_i \lambda_i e_i.$$

Since $\lambda_i \mu_i = \mu_i \lambda_i$ (multiplication of real numbers is commutative), we get

$$(ST)(e_i) = (TS)(e_i) \quad \text{for every } i = 1, \dots, n.$$

Two linear maps that agree on a basis of V are equal. Since ST and TS agree on the basis $e_1, \dots, e_n$, we conclude

$$ST = TS. \qquad \blacksquare$$

Problem 4

Let $V = M_2(\mathbb{R})$, and define $B : V \times V \to \mathbb{R}$ by $B(X, Y) = \operatorname{trace}(XY)$. We show B is a symmetric bilinear form.

Bilinearity

Linearity in the first slot. Let $X_1, X_2, Y \in V$ and $a, b \in \mathbb{R}$. Using distributivity of matrix multiplication over addition, distributivity of scalar multiplication, and linearity of the trace,

$$B(aX_1 + bX_2, Y) = \operatorname{trace}\big((aX_1 + bX_2)Y\big) = \operatorname{trace}(aX_1Y + bX_2Y) = a \operatorname{trace}(X_1Y) + b \operatorname{trace}(X_2Y),$$

so

$$B(aX_1 + bX_2, Y) = aB(X_1, Y) + bB(X_2, Y).$$

Linearity in the second slot. Similarly, for $X, Y_1, Y_2 \in V$ and $a, b \in \mathbb{R}$,

$$B(X, aY_1 + bY_2) = \operatorname{trace}\big(X(aY_1 + bY_2)\big) = \operatorname{trace}(aXY_1 + bXY_2) = a \operatorname{trace}(XY_1) + b \operatorname{trace}(XY_2),$$

so

$$B(X, aY_1 + bY_2) = aB(X, Y_1) + bB(X, Y_2).$$

Hence B is linear in each argument, i.e. B is bilinear.

Symmetry

The trace is invariant under cyclic permutation of a product, so in particular $\operatorname{trace}(XY) = \operatorname{trace}(YX)$ for all $X, Y \in M_2(\mathbb{R})$ (this is a general property of trace which we are permitted to use freely). Therefore

$$B(X, Y) = \operatorname{trace}(XY) = \operatorname{trace}(YX) = B(Y, X)$$

for all $X, Y \in V$, so B is symmetric.

Since B is bilinear and symmetric, B is a symmetric bilinear form on V. $\blacksquare$

Problem 5

Let $n \geq 1$ be an integer and let S_n be the set of bijective functions $\{1, \ldots, n\} \to \{1, \ldots, n\}$, with operation given by function composition, $(\sigma * \tau)(x) = \sigma(\tau(x))$. We verify the group axioms.

Well-definedness / closure

If $\sigma, \tau \in S_n$ are bijections $\{1, \ldots, n\} \to \{1, \ldots, n\}$, then $\sigma \circ \tau$ is also a bijection $\{1, \ldots, n\} \to \{1, \ldots, n\}$:

- *Injective:* if $(\sigma \circ \tau)(x) = (\sigma \circ \tau)(y)$, then $\sigma(\tau(x)) = \sigma(\tau(y))$; since σ is injective, $\tau(x) = \tau(y)$; since τ is injective, $x = y$.
- *Surjective:* let $z \in \{1, \ldots, n\}$. Since σ is surjective, there is y with $\sigma(y) = z$; since τ is surjective, there is x with $\tau(x) = y$. Then $(\sigma \circ \tau)(x) = \sigma(\tau(x)) = \sigma(y) = z$.

So $\sigma \circ \tau \in S_n$, and composition is a well-defined operation $S_n \times S_n \to S_n$.

(i) Associativity

For any $\sigma, \tau, \rho \in S_n$ and any $x \in \{1, \ldots, n\}$,

$$\big((\sigma \circ \tau) \circ \rho\big)(x) = (\sigma \circ \tau)(\rho(x)) = \sigma\big(\tau(\rho(x))\big) = \sigma\big((\tau \circ \rho)(x)\big) = \big(\sigma \circ (\tau \circ \rho)\big)(x).$$

Since this holds for all x, $(\sigma \circ \tau) \circ \rho = \sigma \circ (\tau \circ \rho)$. So composition is associative.

(ii) Identity element

Let $\mathrm{id} : \{1, \ldots, n\} \to \{1, \ldots, n\}$ be defined by $\mathrm{id}(x) = x$. This is clearly a bijection, so $\mathrm{id} \in S_n$. For any $\sigma \in S_n$ and any x,

$$(\mathrm{id} \circ \sigma)(x) = \mathrm{id}(\sigma(x)) = \sigma(x), \qquad (\sigma \circ \mathrm{id})(x) = \sigma(\mathrm{id}(x)) = \sigma(x).$$

Hence $\mathrm{id} \circ \sigma = \sigma \circ \mathrm{id} = \sigma$ for all $\sigma \in S_n$, so id is an identity element for S_n.

(iii) Inverses

Let $\sigma \in S_n$. Since σ is a bijection, it has a (set-theoretic) inverse function $\sigma^{-1} : \{1, \ldots, n\} \to \{1, \ldots, n\}$, defined by: $\sigma^{-1}(y)$ is the unique x with $\sigma(x) = y$. This σ^{-1} is itself a bijection (its inverse is σ), so $\sigma^{-1} \in S_n$. By definition of the inverse function,

$$(\sigma^{-1} \circ \sigma)(x) = \sigma^{-1}(\sigma(x)) = x = \mathrm{id}(x), \qquad (\sigma \circ \sigma^{-1})(y) = \sigma(\sigma^{-1}(y)) = y = \mathrm{id}(y)$$

for all $x, y \in \{1, \ldots, n\}$. So $\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = \mathrm{id}$.

Conclusion

Since composition is a well-defined, associative binary operation on S_n with an identity element id and every element has an inverse in S_n, S_n together with function composition is a group. ■

Problem 2

Let V be a finite dimensional inner product space and $T \in \mathcal{L}(V)$. We prove

$$T = 0 \iff T^* = 0 \iff T^*T = 0 \iff TT^* = 0.$$

Throughout we use the defining property of the adjoint: $\langle Tv, w\rangle = \langle v, T^*w\rangle$ for all $v, w \in V$, together with $(T^*)^* = T$.

Claim A: $T = 0 \iff T^* = 0$

Suppose $T = 0$. Then for all $v, w \in V$,

$$\langle v, T^*w\rangle = \langle Tv, w\rangle = \langle 0, w\rangle = 0.$$

Fix $w \in V$ and take $v = T^*w \in V$: we get $\langle T^*w, T^*w\rangle = 0$, so $\|T^*w\|^2 = 0$, hence $T^*w = 0$. Since w was arbitrary, $T^* = 0$.

Conversely, if $T^* = 0$, apply the implication just proved to the operator $T^* \in \mathcal{L}(V)$ in place of T: it gives $(T^*)^* = 0$. Since $(T^*)^* = T$, this says $T = 0$.

Hence $T = 0 \iff T^* = 0$.

Claim B: $T = 0 \iff T^*T = 0$

If $T = 0$ then clearly $T^*T = 0$.

Conversely, suppose $T^*T = 0$. Then for every $v \in V$,

$$0 = \langle (T^*T)v, v\rangle = \langle Tv, Tv\rangle = \|Tv\|^2,$$

using the defining property of the adjoint with $w = Tv$. So $\|Tv\| = 0$, i.e. $Tv = 0$, for every $v \in V$. Hence $T = 0$.

Claim C: $T^* = 0 \iff TT^* = 0$

Apply Claim B with T replaced by $T^* \in \mathcal{L}(V)$:

$$T^* = 0 \iff (T^*)^*(T^*) = 0.$$

Since $(T^*)^* = T$, the right-hand side is exactly $TT^* = 0$. So $T^* = 0 \iff TT^* = 0$. ■

Problem 3

Let V be a finite dimensional inner product space, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbf{F}$ (where $\mathbf{F} = \mathbb{R}$ or $\mathbf{C}$). We prove

$$\lambda \text{ is an eigenvalue of } T \iff \overline{\lambda} \text{ is an eigenvalue of } T^*.$$

We give a direct proof using only the defining property of the adjoint, $\langle Tv, w\rangle = \langle v, T^*w\rangle$ for all $v, w \in V$, together with conjugate-linearity of the inner product in the second slot: $\langle v, cw\rangle = \overline{c}\,\langle v, w\rangle$ for $c \in \mathbf{F}$.

Reduction by symmetry (WLOG). It suffices to prove the single implication

$$\lambda \text{ eigenvalue of } T \implies \overline{\lambda} \text{ eigenvalue of } T^*. \qquad (*)$$

Indeed, $(*)$ is a statement that holds for *every* operator on V and *every* scalar in $\mathbf{F}$ — the argument below never uses any special feature of T or λ. So we may apply $(*)$ with T^* in place of T and $\overline{\lambda}$ in place of λ (recalling $(T^*)^* = T$ and $\overline{\overline{\lambda}} = \lambda$), which gives the reverse implication for free:

$$\overline{\lambda} \text{ eigenvalue of } T^* \implies \overline{\overline{\lambda}} \text{ eigenvalue of } (T^*)^* \iff \lambda \text{ eigenvalue of } T.$$

So proving $(*)$ once proves both directions simultaneously, and it remains only to establish $(*)$.

Proof of $(*)$

Suppose λ is an eigenvalue of T. Then $T - \lambda I$ is not injective, so, since V is finite dimensional, $T - \lambda I$ is also not surjective (by rank–nullity: injective $\iff$ surjective on a finite dimensional space). Hence

$$U := \operatorname{range}(T - \lambda I)$$

is a proper subspace of V. Since $V = U \oplus U^\perp$ for any subspace U of a finite dimensional inner product space, and $U \neq V$, we must have $U^\perp \neq \{0\}$. Choose a nonzero $w \in U^\perp$.

By definition of $U^\perp$, $\langle (T - \lambda I)v, w\rangle = 0$ for every $v \in V$, i.e.

$$\langle Tv, w\rangle = \lambda\langle v, w\rangle \qquad \text{for all } v \in V.$$

Now rewrite both sides using the adjoint and conjugate-linearity:

$$\langle Tv, w\rangle = \langle v, T^*w\rangle, \qquad \lambda\langle v, w\rangle = \langle v, \overline{\lambda}\, w\rangle$$

(the second equality holds because $\langle v, \overline{\lambda}w\rangle = \overline{\overline{\lambda}}\,\langle v, w\rangle = \lambda\langle v, w\rangle$). Combining,

$$\langle v, T^*w\rangle = \langle v, \overline{\lambda}\, w\rangle \qquad \text{for all } v \in V,$$

i.e. $\langle v,\, T^*w - \overline{\lambda}w\rangle = 0$ for all $v \in V$. Taking $v = T^*w - \overline{\lambda}w$ gives $\|T^*w - \overline{\lambda}w\|^2 = 0$, so

$$T^*w = \overline{\lambda}w.$$

Since $w \neq 0$, this shows $\overline{\lambda}$ is an eigenvalue of T^* (with eigenvector w), proving $(*)$.

Conclusion

By the symmetry reduction above, $(*)$ alone gives

$$\lambda \text{ is an eigenvalue of } T \iff \overline{\lambda} \text{ is an eigenvalue of } T^*. \qquad \blacksquare$$